

AI Learning Roadmap (Beginner to Production Level)

Start small → build understanding → increase complexity → reach production level. This roadmap helps students become industry-ready AI engineers.

LEVEL 1: BASIC NLP

Project 1: Text Cleaner + Analyzer

- 1 Remove stopwords
- 2 Tokenization
- 3 Word frequency counter

Skills: Python basics, NLP fundamentals

Project 2: PDF Reader + Search

- 1 Read PDF
- 2 Search keyword
- 3 Return matching paragraph

Skills: PDF processing, string matching

LEVEL 2: YOUR FIRST AI

Project 3: TF-IDF Search Engine

- 1 Train TF-IDF
- 2 Ask questions
- 3 Return best answer

Skills: Vectorization, cosine similarity

Project 4: PDF Q&A; System

- 1 Chunk PDF
- 2 Convert to vectors
- 3 Retrieve best chunk

Skills: First AI system design

LEVEL 3: DEEP LEARNING

Project 5: Text Classification

- 1 Spam detection
- 2 Topic classification

Skills: PyTorch, training loop

Project 6: Text Generator (LSTM)

- 1 Train sequence model
- 2 Generate text

Skills: Deep learning basics

LEVEL 4: LLM FUNDAMENTALS

Project 7: Mini GPT

- 1 Tokenizer
- 2 Attention mechanism
- 3 Transformer

Skills: LLM architecture

LEVEL 5: REAL-WORLD SYSTEM

Project 8: RAG System

- 1 PDF upload
- 2 Vector search
- 3 Context-based answers

Skills: System design

LEVEL 6: PRODUCTION

Project 9: AI Backend API

- 1 /upload
- 2 /train
- 3 /ask

Skills: FastAPI, backend

Project 10: Full SaaS AI App

- 1 UI
- 2 Database

3 Authentication

Skills: Full-stack AI system

Final Outcome

- 1 NLP understanding
- 2 Machine Learning
- 3 Deep Learning
- 4 LLM basics
- 5 RAG systems
- 6 Backend APIs
- 7 Production deployment

Roadmap Flow

Text → NLP → ML → Deep Learning → LLM → RAG → API → Production